Auto Connect

ISP Automation System

Pranjal Rajesh Upadhyay
PRN: 1032233275
Roll No: 39 | Panel: B
Prof. Suhas Joshi

Console-Based Java Project

Mini Project using JAVA Programming

- OOP
- Role-Based Auth
- Password Security
- Admin Dashboard

Project Overview

What is AutoConnect?

A terminal-based ISP Management System built using Java and OOP principles.

v1.0 Single flat menu, basic features, no authentication.

v2.0 Dual-role architecture, secure login, admin controls, complaint tracking.

Core Entities

Customer	Name, Phone, Email, Password, Status
Admin	Name, Email, Password (hardcoded)
Plan	ID, Name, Price (Basic/Standard/Premium)
Bill	ID, Customer ID, Amount, Status
Complaint	ID, Customer ID, Issue, Status, Technician

System Architecture

MAIN CLASS

AutoConnect_Updated

Console UI loop
Role selection + input handling

SERVICE LAYER

ISPService

Business logic
All operations & data management

MODEL LAYER

Classes

Customer · Admin · Plan
Bill · Complaint

Supporting Enums:

ConnectionStatus ACTIVE | SUSPENDED

BillStatus PAID | UNPAID

Role-Based Access & Authentication

ADMIN

Auth: Email + Password (hardcoded)

- View all customers, bills, complaints
- Assign technician to complaint
- Update complaint status
- Suspend / Activate customer
- View admin profile

SECURITY THREAT DETECTED! Unauthorized access attempt. Access Denied.

CUSTOMER

Auth: Email + Password (registered)

- Register with password strength check
- View own profile & plans
- Generate & pay bills
- Raise & track complaints
- Change password (STRONG only)

Invalid credentials! Please register first.

Password Strength Checker – PasswordUtil

5 Scoring Criteria (1 point each):

1 Length >= 8	2 Uppercase [A-Z]	3 Lowercase [a-z]	4 Digit [0-9]	5 Special char

Strength Levels:

WEAK	MEDIUM	STRONG
Score 0–2	Score 3–4	Score 5
e.g. "karan123"	e.g. "Rahul123"	e.g. "Aisha@1234"

WEAK → Rejected with tip message **MEDIUM** → User chooses: keep or re-enter **STRONG** → Confirm password step

Key Enhancements – v1.0 vs v2.0

Feature	v1.0 (Old)	v2.0 (New)
User Roles	Single flat menu	Admin + Customer (separate flows)
Authentication	None	Email + Password login for both roles
Registration	Name + Phone only	Name, Phone, Email, Password
Complaint View	Shows ALL complaints	Filtered by logged-in customer ID
Technician	Not present	Assignable + status tracked
Suspend/Activate	suspend() unused in UI	Admin dashboard toggle
Change Password	Not present	Strength-checked update

Admin Dashboard

1 **View Yourself**

Displays admin name, email, role: ADMINISTRATOR

2 **View All Customers**

Lists all registered customers with status

3 **View All Bills**

Shows all bills with amount and payment status

4 **View All Complaints**

Lists all complaints with technician info

5 **Assign Technician**

Sets technician on a complaint → IN PROGRESS

6 **Update Complaint Status**

RESOLVED / IN PROGRESS / WAITING

7 **Suspend / Activate Customer**

Toggles ACTIVE ↔ SUSPENDED for any customer

8 **Logout**

Returns to main role-selection menu

Complaint Lifecycle & Technician Workflow

Customer raises issue | Technician assigned | Pending info / parts | Issue fixed & closed

Complaint Class Fields

id : *int* Unique complaint ID

status : *String* OPEN / IN PROGRESS / WAITING / RESOLVED

cid : *int* Customer ID (owner)

technician : *String* "NOT ASSIGNED" by default → updated by admin

technician : *String* "NOT ASSIGNED" by default → updated by admin via assignTechnician()

Sample Console Output

Registration – WEAK password rejected

```
Who are you? 1.Admin  2.Customer  3.Exit
> 2
1.Register  2.Login  3.Back
> 1
Name: Karan   Phone: 9012345678
Email: karan.tech@gmail.com
Password: karan123
Password Strength: WEAK
Password is too weak!
Tip: Use uppercase, lowercase,
     number & special char (min 8).
Password: Karan@Tech9
Password Strength: STRONG
Confirm Password: Karan@Tech9
Registered Successfully!
```

Admin – Security Warning & Login

```
Who are you? 1.Admin  2.Customer  3.Exit
> 1
Admin Email: hacker@x.com
Admin Password: wrongpass
SECURITY THREAT DETECTED!
Unauthorized access attempt.
Access Denied.
---
Admin Email: rajeshu0707@gmail.com
Admin Password: admin@01
Welcome Admin Rajesh!

===== Admin Dashboard =====
1. View Yourself
2. View All Customers ...
```

Key Takeaways

Dual-Role Architecture	Admin & Customer with separate authenticated flows
Password Security	PasswordUtil: WEAK / MEDIUM / STRONG + confirm step
Admin Controls	8-option dashboard: view, manage, suspend, assign
Complaint Tracking	Technician field + 4-state status lifecycle
Filtered Data Access	Customers see only their own complaints & bills